

Course Title	Tests of hypothesis with JASP software				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Familiarity with common data formats, awareness of hypothesis testing concepts including null and alternate hypothesis, significance levels and p-values.				
Course Summary	Cover sampling, probability distributions, and mediation/moderation analysis, introducing JASP software for correlation, t-tests, and ANOVA. Equip students with skills for hypothesis testing, normal distribution properties, and psychological research analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	Explain why sampling is used in statistical analysis.	U	C	Instructor-created exams / Quiz
CO 2	Describe and explain various sampling techniques such as simple random sampling, stratified sampling.	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO 3	Introduce mediation and moderation analysis concepts .	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO 4	Define what non parametric tests are and explain how they differ from parametric tests and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Instructor-created exams / Home Assignments
CO 5	Define correlation and explain its significance in statistical analysis and critically evaluate ethical implications of statistical methods aligning with human values.	R	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO 6	Apply JASP software for hypothesis testing and analysis.	Ap	P	Viva Voce/ Instructor-created exams

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks 70
I	Sampling methods		10	15
	1	Introduction	1	
	2	Types of sampling	1	
	3	Purposive sampling	2	
	4	Random sampling	2	
	5	Simple sampling	2	
	6	Stratified sampling	2	
	Sections from References: Unit 1: 12.1 Ref[2] Unit 2:12.2 Ref[2] Unit 3: 1 2.2.1 Ref[2] Unit 4: 12.2.2 Ref[2] Unit 5: 12.2.3 Ref[2] Unit 6: 12.2.4 Ref[2]			
II	An introduction to Mediation analysis		9	15
	7	The Simple Mediation Model	2	
	8	Estimation of the Direct, Indirect, and Total Effects of X- Brief concept	2	

	9	Concept of confounding and causal order	2	
	10	Conditional and Unconditional Effects	3	
	Sections from References: Unit 7: 3.1 Ref[3] Unit 8: 3.2 Ref[3] Unit 9: 4.2 Ref[3] Unit 10: 7.1 Ref[3]			
III	Introduction to Non parametric tests		14	20
	11	Non-parametric Methods	2	
	12	Advantages and Disadvantages of Non parametric Methods over parametric methods	2	
	13	Sign test	3	
	14	Median test	2	
	15	Mann Whitney Wilcoxon U test	2	
	16	Wald-Wolfowitz Run Test	3	
	Sections from References: Unit 11: 16.8 Ref[2] Unit 12 16.8.1 Ref[2] Unit 13:16.8.3 Ref[2] Unit 14: 16.8.5 Ref[2] Unit 15:16.8.7 Ref[2] Unit 16: 16.8.3 Ref[2]			
IV	Correlation and test in JASP software		12	20
	17	The one-sample z-test .	2	
	18	The one-sample t-test .	2	
	19	The independent samples t-test	3	
	20	The paired-samples t-test	2	

	21	Correlations	2	
	22	Scatter plots	1	
	Sections from References: Unit 17: 10.1 Ref[4] Unit 18: 10.2 Ref[4] Unit 19: 10.3 Ref[4] Unit 20: 10.5 Ref[4] Unit 21:11.1 Ref[4] Unit 22: 11.2 Ref[4]			
V	PRACTICUM		30	
	Do practice problems using JASP software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Checking the normality of a sample		
	2	Testing non normal data with Wilcoxon tests		
	3	Reporting the results of a hypothesis test		
	4	Making decisions		
	5	p value of a test		
	6	Running hypothesis test in practice		
	7	Discussion on various examples of population		
	8	Discussion on simple random sampling		
	Sections from References: Unit 1: 10.8 Ref[4] Unit 2:10.9 Ref[4] Unit 3: 8.6 Ref[4]			

Unit 4: 8.4 Ref[4]		
Unit 5: 8.5 Ref[4]		
Unit 6: 8.7 Ref[4]		
Unit 7: 7.1.1 Ref[4]		
Unit 8:7.1.2 Ref[4]		
Books and References: 1. S.P Gupta (2021),Statistical Methods 46 th Edition 2. Gupta, S.C. and Kapoor, V.K. (1997) Fundamentals of Mathematical Statistics. Sultan Chand and Sons, New Delhi 3. Hayes, A.F. (2017) Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach. Guilford Press, New York 4. Navarro, D.J., Foxcroft, D.R., & Faulkenberry, T.J. (2019). Learning Statistics with JASP: A Tutorial for Psychology Students and Other Beginners. (Version		

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	2	-	-	-	-	1	1	-	-	-	-
CO 2	-	-	-	-	-	3	2	1	-	-	-	-
CO 3	-	2	-	3	2	-	2	2	2	-	-	-
CO 4	-	-	3	-	-	3	3	3	2	-	3	-
CO 5	3	-	-	-	-	2	1	2	-	-	-	3
CO 6	-	-	3	-	-	-	2	3	-	3	-	-

Correlation Levels:

Lev	Correlation
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el	
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignm ent	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA1MN107 (P)
Course Title	Basic statistics

Type of Course	Minor				
Semester	I				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of data, variables, charts and graphs. Basic computer skills				
Course Summary	To provide students with a fundamental understanding of life science data and statistical methods for its analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate proficiency in collecting life science data and understanding the distinction between population and sample	U	C	Instructor-created exams / Quiz
CO2	Identify and analyze sampling and non-sampling errors, determine appropriate sample sizes, and recognize different variables and observational units within life science datasets.	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Apply frequency distributions and plotting techniques to visualize life science data effectively and analyze data to help entrepreneurial decisions using critical thinking skills.	Ap	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Explore relationships between variables within life science datasets, employing statistical methods to quantify and analyze these relationships and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Home Assignments/ Instructor-created exams
CO5	Define key terms in probability, including events, outcomes and sample spaces.	R	P	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Develop a basic understanding of how to do measures of central tendency and dispersion using spread sheet.	Ap	C	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)				

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)
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COURSE CONTENT

Module	Unit	Content	Hrs (45 +30)	Marks (70)
1	Collection of Data and Sampling		10	20
	1	Examples of Life Science data	1	
	2	Collection of data- Primary and secondary data,	1	
	3	Population and Sample, Census and Sampling	1	
	4	Advantages and Limitations of Sampling	1	
	5	Simple Random Sampling (Concept and Method of Selection)	2	
	6	Stratified Random Sampling	2	
	7	Systematic Random Sampling	1	
	8	Sampling and Non Sampling Errors	1	
	Sections from References: Unit 1: 1.1[Ref 1] Unit 2:2.2 [Ref 2] Unit 3: 15.2,15.3,15.6 [Ref 2] Unit 4: 15.6,15.7[Ref 2] Unit 5:15.11,15.11.1 [Ref 2] Unit 6: 15.12,15.12.1 [Ref 2] Unit 7: 15.13 [Ref 2] Unit 8:15.9.1[Ref 2]			
2	Frequency Distribution and Descriptive Statistics		12	10
	9	Frequency Distribution	2	
	10	Cumulative Frequency distribution	2	
	11	Diagrammatic Representations	4	
	12	Graphical Representation of data	4	
	Sections from References: Unit 9: 3.3[Ref 2]			